

Medication Virtual Assistant

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1 Use Case

- **Problem:** Patients may consume medicine combinations that can be dangerous for their health. Patients face issues finding a reliable or accurate source that can guide them, which medicine to take during particular symptoms.
- **Solution:** A medical virtual assistant will help them determine what medicines to consume on the basis of their medical history. Unlike other chatbots, this will retrieve data from trustable sources(web based search) and from the patient's medical history.
- **Target Users:** Patients who need personal guidance on what medications they should take according to particular symptoms. Patients who want quick guidance on what medicine to take based on their symptoms and medical history. People with long-term illnesses or who take multiple medicines. Elderly patients and caregivers who manage daily medications. Users with allergies or past bad reactions to medicines. People in areas with limited access to doctors/pharmacists who need trusted basic advice and "when to see a doctor" warnings.

2 Planned User Queries

Database Search:

1. What is Metformin used for?
2. What are the side effects of Aspirin?
3. What is the recommended dosage for Ibuprofen?

Drug Interactions:

9. Can I take Aspirin with Warfarin?
10. Is it safe to take Ibuprofen and Aspirin together?
11. Can I drink alcohol while taking Metformin?

Web Search/Latest Info:

12. Were there any recent recalls for Metformin?
13. What's the latest news about Aspirin safety?
14. Are there any new warnings for Lisinopril in 2026?

3 Data sources

- **OpenFDA API** - consists of drug labels, medication info, current status of drug adverse event reports
- **RxNorm/RxNav** - consists of info about how two drugs interact with each other.
- **Synthea** - Synthetic data creation tool that creates synthetic patient history data so privacy is not compromised.
- **Dataset** - 50-100 medications, 50+ drug interactions, patient records.

4 Tools

1. **Database Search Tool** - Provides info based on uses, dosage and side effects of medicines.
2. **Drug Interaction Tool** - Provides info about how various drugs will interact with each other.
3. **Web Search Tool** - Provides info about recent events/latest news about drugs.

5 Models

- **Model 1: Meta Llama-3-8B** - 8B parameters, Expected: Higher accuracy, slower response
- **Model 2: Microsoft Phi-3.5-mini-4k-instruct** - 3.8B parameters, lightweight and fast, Expected: Faster response, acceptable accuracy

6 Technical Approach

- **Architecture:** User query - LLMs(Decides the appropriate tool) - Rag Retrieval(executes retrieval query and extract the data) - LLM(Rag Generation uses prompting techniques, Retrieved data) - Final Response to user.

Key Techniques

- **RAG:** Helps in retrieving the data from database and generating the response.
- **Function calling:** it decides the appropriate tool on the basis of task
- **Prompting:** Chain of thought, few shot learning, self reflection.
- **Prompt caching:** helps to optimize the speed by not recomputing the task.

6.1 Evaluation Metrics

- **First-Turn Resolution (FTR)** = percentage solved in the first reply
- **Turns-to-Resolution** = average turns needed to solve
- **Abandonment Rate** = percentage of users who leave before solving
- **Repeat-Question Rate** = percentage of users who ask the same thing again

7 Expected Deliverables

- **Medication Database:** will consist of 50 to 100 different types of drugs
- **Model comparison Report:** comparison on the basis of speed and accuracy of two different llm models.
- **User Interface:** Gradio/streamlit
- **Demo video + final report and presentation**

8 References

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